

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA -2
Academic year	I – III Semester(2025-26)
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Branch	AI&DS
Course Outcome	1

Case Study Topic : QuickCart Warehouse — package-tracking index system (AVL)

The QuickCart logistics warehouse maintains a package dictionary indexed by 7-digit Package IDs. During the morning dispatch cycle, packages are scanned in truck-loading order. Since labels are generated sequentially at packaging time, the IDs arrive in nearly sorted order.

Word Done Summary:

The QuickCart Warehouse Package-Tracking Index System was developed to efficiently manage and track packages within a warehouse environment using advanced data structures and indexing techniques. The project focused on organizing package information, including tracking IDs, storage locations, shipment status, and delivery updates, to enable fast searching and retrieval of records. An optimized indexing mechanism was implemented to reduce lookup time and improve warehouse operations. The system also supports real-time package status updates, inventory monitoring, and efficient package movement tracking. By streamlining package management and providing quick access to tracking information, the solution enhances operational efficiency, minimizes delays, improves accuracy in package handling, and supports better decision-making for warehouse administrators and logistics teams.

Implementation Details:

1. Designed a database to store package details, tracking IDs, locations, and shipment status.
2. Implemented an indexing system for fast package search and retrieval.
3. Developed modules for package registration and tracking ID generation.
4. Enabled real-time updates of package movement and delivery status.
5. Integrated warehouse location mapping for efficient package storage management.
6. Created search and monitoring features to track packages instantly.

7. Developed an admin dashboard for managing inventory, tracking records, and warehouse operations.

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● b.deepak@BDeepaks-MacBook-Air DSA project % javac AVLNode.java
● b.deepak@BDeepaks-MacBook-Air DSA project % java AVLNode
  Root key: 10
○ b.deepak@BDeepaks-MacBook-Air DSA project % █
```

GitHub Link:

<https://github.com/bdeepakg-boop/DSA-Project/upload/main>

Student Signature	Faculty Signature